Linear Phase FIR Filters

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Key Concepts

1) One of the big advantages of FIR filters is that they can be guaranteed to have generalized linear phase response, and thus introduce no phase distortion.

2) Requiring linear phase places constraints on the filter coefficients.

3) There are four different categories of linear phase FIR filters. Let M be the filter order and $h[n], 0 \leq n \leq M$ be the filter coefficients.

    a) Type I filters have M even and even symmetry in the coefficients $h[n] = h[M - n]$.

i) No constraints on the magnitude response.

ii) The group delay is $M/2$.

b) Type II filters have M odd and even symmetry in the coefficients $h[n] = h[M - n]$.

i) The frequency response at $\omega = \pi$ is always zero.

ii) Cannot be used to design high pass filters.

iii) The group delay is $M/2$.

c) Type III filters have M even and odd symmetry in the coefficients $h[n] = -h[M - n]$.

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i) The frequency response at $\omega = 0$ and $\omega = \pi$ is always zero.

ii) Cannot be used to design low pass or high pass filters.

iii) The group delay is $M/2$.

d) Type IV filters have M odd and odd symmetry in the coefficients $h[n] = -h[M - n]$.

i) The frequency response at $\omega = 0$ is always zero.

ii) Cannot be used to design low pass filters.

iii) The group delay is $M/2$.

4) Zeros not on the unit circle must occur in complex conjugate reciprocal pairs.

Video Screenshots

FIR Filter

$$H(e^{j\omega}) = \sum_{k=0}^{M} b_k e^{-jk\omega} = \sum_{k=0}^{M} h[k] e^{-jk\omega}$$

Generalized linear phase

$$H(e^{j\omega}) = A(e^{j\omega}) e^{-j\alpha\omega + j\beta}$$

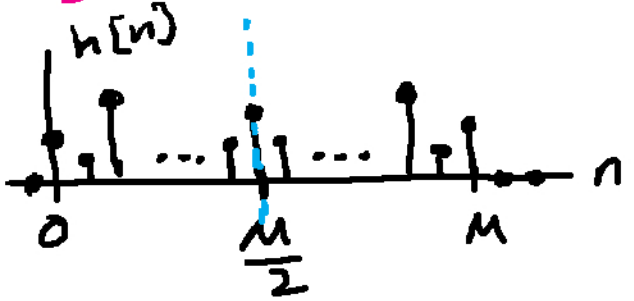
$\Rightarrow$ constant group delay

Four types of FIR filters w. generalized linear phase

M even or M odd, even or odd symmetry

Type I : M even, even symmetry

$$h[n] = h[M-n], \quad 0 \le n \le M$$



$$H(e^{j\omega}) = \sum_{k=0}^{M} h[k] e^{-jk\omega}$$

$$= e^{-jM/2\,\omega}\left(h[0]e^{j\frac{M}{2}\omega} + h[M]e^{-j\frac{M}{2}\omega} + h[1]e^{j(\frac{M}{2}-1)\omega} + h[M-1]e^{-j(\frac{M}{2}-1)\omega} + \cdots + h[M/2]e^{j0\omega} \right)$$

$$= e^{-j\frac{M}{2}\omega}\left(2h[0]\cos(\tfrac{M}{2}\omega) + 2h[1]\cos((\tfrac{M}{2}-1)\omega) + \cdots + h[\tfrac{M}{2}] \right)$$

$$= e^{-j\frac{M}{2}\omega} \sum_{k=0}^{M/2} a[k]\cos(\omega k) \quad ; \qquad a[0] = h[M/2]$$

$$a[l] = 2h[\tfrac{M}{2}-l], \quad l = 1, \cdots, \tfrac{M}{2}$$

$$H(e^{j\omega}) = e^{-jM/2\omega} \underbrace{\sum_{k=0}^{\frac{M}{2}} a[k]\cos(\omega k)}_{\text{real}}$$

Generalized Linear Phase

$$H(e^{j\omega}) = A(e^{j\omega})\, e^{-j\alpha\omega + j\beta}$$

$$\Rightarrow A(e^{j\omega}) = \sum_{k=0}^{M/2} a[k]\cos(\omega k)$$

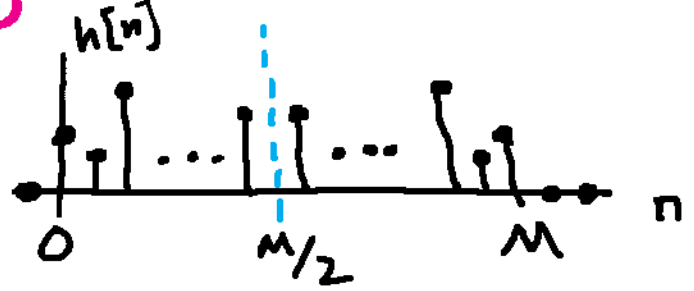
$$\alpha = M/2$$

Group delay $M/2$

Type II : M odd, even symmetry

4:49

$$h[n] = h[M-n], \quad 0 \le n \le M$$



$$\Downarrow$$

$$H(e^{j\omega}) = e^{-j\omega(M/2)} \sum_{k=1}^{(M+1)/2} b[k] \cos\left[\omega(k-1/2)\right]$$

$$b[k] = 2h[(M+1)/2 - k], \quad k = 1, 2, \ldots, (M+1)/2$$

$$\cos\left[\pi(k-1/2)\right] = 0 \quad \text{for all integer } k$$

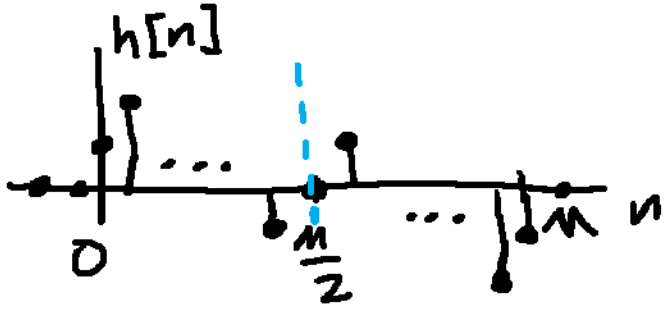
$$\Rightarrow H(e^{j\pi}) = 0 \qquad (\text{zero at } z = -1)$$

no highpass filters

Type III: M even, odd symmetry

6:50

$$h[n] = -h[M-n]$$



$$H(e^{j\omega}) = j e^{-j\omega M/2} \sum_{k=1}^{M/2} c[k] \sin \omega k$$

$$c[k] = 2h[M/2 - k]$$

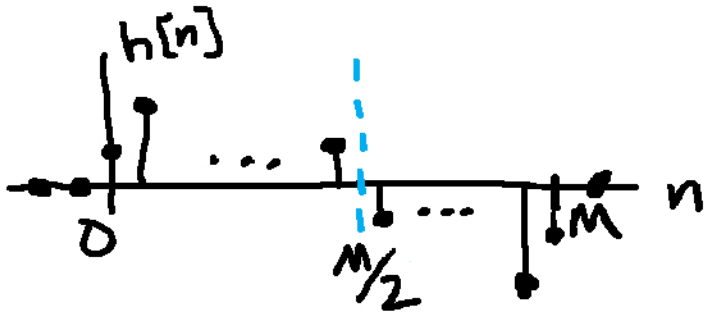
$\sin(0k) = 0$, $\sin(\pi k) = 0$ forall integer k

$\Rightarrow$ $H(e^{j0}) = 0$, $H(e^{j\pi}) = 0$ (zeros at $z = \pm 1$)

no lowpass or high pass filters

Type IV: M odd, odd symmetry

$$h[n] = -h[M-n]$$



$$H(e^{j\omega}) = j e^{-j\omega M/2} \sum_{k=1}^{(M+1)/2} d[k] \sin[\omega(k-1/2)]$$

$$d[k] = 2\, h[(M+1)/2 - k], \quad k = 1, 2, \cdots (M+1)/2$$

$$\sin[0(k-1/2)] = 0 \quad \text{for all } k$$

$$\Rightarrow H(e^{j0}) = 0 \qquad (\text{zero at } z = 1)$$

no lowpass filters

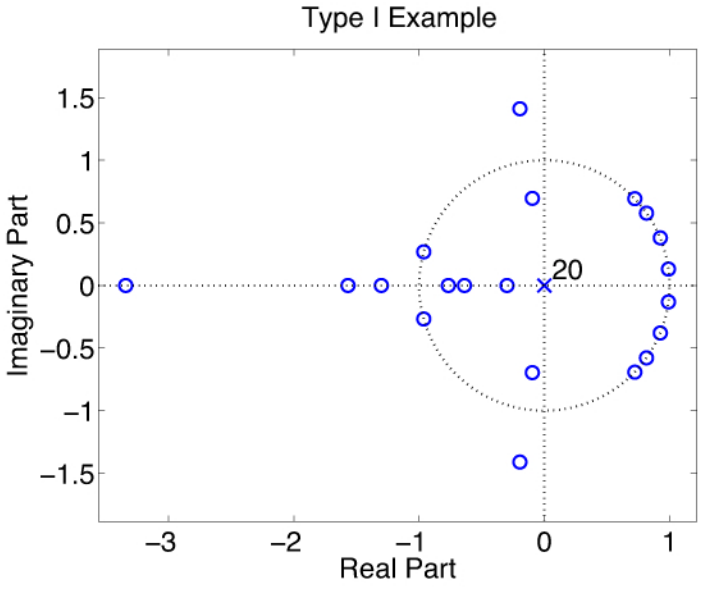
Type I + II (even symmetry)

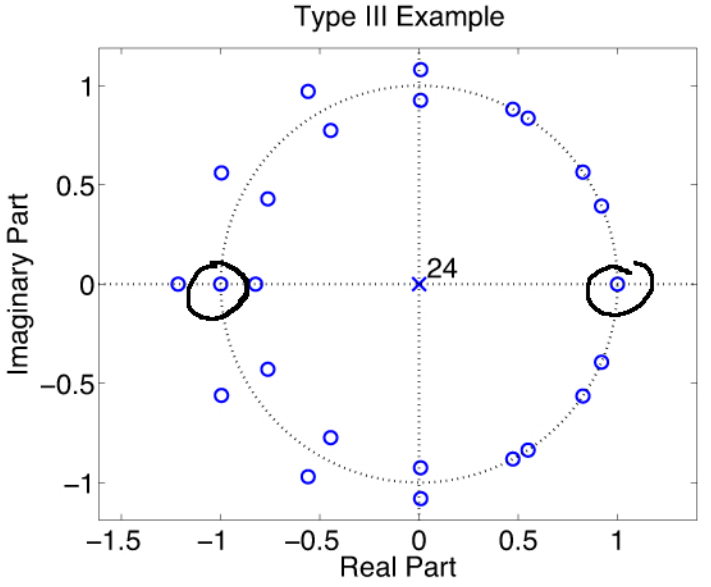
$$H(z) = z^{-M} H(z^{-1})$$

Type III + IV (odd symmetry)

$$H(z) = -z^{-M} H(z^{-1})$$

zeros not on $|z|=1$ are in conjugate reciprocal pairs





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